SoftFuse-KGC: Comment 2 Calibration Experiments

Revision working draft

5 November 2026

Completed experiment; standalone insertion draft. This document assembles the new methods and results for review. It is not the integrated manuscript or supplementary submission. Historical metrics must be reconciled before replacing their tables with the standardized results below.

1 Automatic calibration study

1.1 Automatic coefficient calibration and transfer analysis

To investigate dependence on dataset-specific manual coefficients, we distinguish calibration of the original support formula from a complementary calibration baseline with shared feature definitions. The original formula retains its dataset-specific evidence features, while a fixed validation search selects the shortcut penalty, contradiction penalty, and rank-prior mixture from a declared grid of 224 configurations per dataset. Selection uses validation MRR@20 only; the selected configuration is frozen before test evaluation.

The shared-feature baseline uses the original rank prior, source-subgraph contact, bounded log degree, bounded log shared-neighbor count, and a negative direct-contact indicator. These features are computed without reference to evaluation labels and use the same definitions and normalization constants across datasets. They are topological descriptors rather than annotations of therapeutic mechanism. An equal-weight baseline is included to separate effects of the shared features from effects of coefficient learning.

For learned calibration, a linear candidate score is fitted using a pairwise logistic objective with L2 regularization. The primary formulation constrains coefficients to be nonnegative because feature signs are specified explicitly; an unconstrained counterpart is also evaluated. The regularization coefficient is selected by five-fold query-group cross-validation within the existing validation split, which serves as a calibration pool. Candidates known to be other positives for the same calibration-training query are excluded as negative pairs. Queries whose gold entity is absent from the fixed top-20 list cannot contribute a positive training pair, but remain in all evaluation denominators.

We additionally perform leave-one-dataset-out coefficient transfer: the target dataset contributes no validation labels to fitting or hyperparameter selection. Coefficients are fitted on the other datasets, with equal total pair weight per source dataset, and regularization is selected using leave-one-source-dataset-out validation. The target graph, structural retriever, and prepared subgraphs remain available. This experiment therefore assesses transfer of calibration coefficients, not zero-shot completion on an unseen knowledge graph. Pairwise source-to-target transfer, feature ablation, calibration-label learning curves, and coefficient perturbation analyses provide complementary diagnostics.

For fuzzy retrieval, a common edge feature set uses rank-band confidence, candidate incidence, query incidence, direct-contact penalties, and bounded degree information, without dataset-specific relation-family priorities. Automatic selection searches a fixed set of coefficient vectors using validation generated-answer accuracy, with adjusted MRR@20 as a secondary criterion. Candidate order and per-query edge budgets are held fixed. A source-only macro-validation criterion provides the corresponding leave-one-dataset-out edge-coefficient selection. All selected settings are recorded before test inference. The source-only restriction applies to this new coefficient-selection step; the original fuzzy candidate order and per-query budgets are inherited from the historical dataset-specific pipeline. Thus the complete graph-input pipeline is not claimed to be free of prior target-dataset calibration.

End-to-end comparisons use one frozen checkpoint per dataset. Candidate-order interventions retain the full source graph, whereas edge-selection interventions retain the original fuzzy input candidate order. Exact generated-answer accuracy, invalid-output rate, and top-candidate-copy rate are reported separately from adjusted MRR@20. Paired query-cluster bootstrap intervals quantify uncertainty; multiple gold rows for the same query are resampled together. These analyses are retrospective checks on previously studied benchmarks, and their exploratory intervals are not adjusted for multiple comparisons.

Table 1: Test candidate MRR@20. Original uses the native soft-support formula; the other columns use identical shared feature definitions. LODO excludes target validation labels from coefficient fitting and regularization selection. These comparisons retain exactly the same top-20 candidate membership.

Dataset	Original	Shared equal	Shared learned	Shared LODO
PrimeKG	0.125326	0.125188	0.125326	0.126104
PharmKG	0.028159	0.031883	0.031876	0.031860
Hetionet	0.095439	0.101269	0.115284	0.113579
DRKG	0.046556	0.061426	0.060496	0.058612
repoDB	0.048107	0.057366	0.064902	0.064902

Table 2: Generated-answer exact accuracy (%) in the candidate-order arm with the full source graph and a fixed checkpoint within each dataset. Accuracy is separate from candidate MRR and adjusted MRR.

Dataset	Test rows	Original	Shared equal	Shared learned	Shared LODO
PrimeKG	500	0.0	0.0	0.0	0.0
PharmKG	500	0.0	0.2	0.2	0.2
Hetionet	100	3.0	1.0	1.0	2.0
DRKG	500	1.4	0.8	2.0	1.8
repoDB	500	0.2	0.8	0.0	0.0

1.2 Results of automatic calibration and coefficient transfer

All five datasets completed the calibration protocol and full-test evaluation. The study includes 1,120 native validation configurations, 20 directed source–target transfers, 100 calibration-label subset runs, and 200 coefficient perturbation runs. Frozen-checkpoint inference evaluates 60 dataset–edge-vector combinations on validation and 55 dataset–variant combinations on test. The same 2,100 validation rows and 2,100 test rows are reused across their respective variants; these are not independent new examples. Query-level caching avoids repeated generation for identical model inputs.

Candidate-level results show that a common calibration feature space can retain or improve the point estimates of the native soft-support formula. However, the equal-weight control already improves several datasets, so comparisons against the original formula combine feature changes with coefficient learning. Against the feature-matched equal-weight control, learned calibration changes MRR by 0.000138, −0.000007, 0.014015, −0.000930, and 0.007536 on PrimeKG, PharmKG, Hetionet, DRKG, and repoDB, respectively. The exploratory paired query-cluster 95% bootstrap interval excludes zero only for repoDB among these five comparisons: [0.000402, 0.015176]. Thus, the results do not establish that learning uniformly outperforms a common fixed weighting.

The source-only LODO coefficients produce MRR point estimates above the native formula on all five datasets, but they do not uniformly improve over shared equal weights. In particular, DRKG shared equal weights yield 0.061426 versus 0.058612 for LODO. The LODO study concerns coefficient transfer within the existing benchmark pipelines and should not be interpreted as zero-shot completion or generalization to unseen diseases.

Learning curves also differ by dataset. On PharmKG, mean test MRR changes from 0.028975 at 10% of calibration query groups to 0.031856 at 100%, with standard deviations 0.004282 and 0.000020 across subset/CV seeds, respectively. PrimeKG is unchanged at 0.125326 over the tested fractions. Hetionet and repoDB do not show monotonic improvement as the calibration fraction increases. These seed-based standard deviations describe calibration variability, not confidence intervals for performance on independent test sets.

End-to-end outcomes are mixed (Table 2). Higher candidate MRR does not ensure higher generated-answer accuracy. For example, repoDB learned calibration improves candidate MRR but yields no exact answers, compared with four correct answers among 500 rows for shared equal weights. The paired accuracy difference is −0.8 percentage points, with an exploratory 95% interval of [−1.702, −0.190] percentage points. On DRKG, learned calibration yields 2.0% exact accuracy versus 0.8% for equal weights; the corresponding interval for the 1.2-point difference is [0.000, 2.469] percentage points. These findings limit claims about improvements in generation.

For edge calibration, the numbers of vectors tied at the best validation accuracy and adjusted MRR are 6/12, 12/12, 12/12, 12/12, and 8/12 for PrimeKG, PharmKG, Hetionet, DRKG, and repoDB, respectively. The prespecified tie rule selects the earliest vector, rather than using test results to break ties. All four compressed-graph variants in Table 3 have the same exact-answer accuracy within each dataset: 0.0%, 0.2%, 3.0%, 1.4%, and 0.4%, respectively. Edge selection therefore changes some graph properties and adjusted ranks without establishing an exact-answer advantage. Equal metrics under the tested vectors do not identify a unique optimum or prove equivalence for other graphs, budgets, or checkpoints.

All results above are retrospective analyses of existing benchmark splits. Bootstrap intervals are exploratory

Table 3: Test adjusted MRR@20 in the graph-selection arm. Candidate order and the compressed-graph budget are fixed per query. Invalid outputs retain the explicit virtual-rank-one convention described in the protocol; these values must not be read as generated-answer accuracy.

Dataset	Original fuzzy	Shared fixed	Shared auto	Edge LODO
PrimeKG	0.074687	0.074687	0.074742	0.074687
PharmKG	0.020971	0.021290	0.021290	0.021290
Hetionet	0.100087	0.100087	0.100087	0.100087
DRKG	0.046285	0.046285	0.046285	0.046285
repoDB	0.037776	0.037795	0.037795	0.038045

and are not adjusted for multiple comparisons. Full configuration traces, coefficients, query-level predictions, feature ablations, transfer matrices, sensitivity analyses, and paired comparisons are provided by the accompanying reproducibility scripts.